

Uniqueness of the Rank-5 Projection in the Quantum Gauge Theory of Time

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Abstract

The Quantum Gauge Theory of Time (QGT) is grounded on a non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ of rank five. While this rank has been derived from spectral arguments in prior work, its uniqueness — the proof that no other rank is compatible with the three foundational requirements of the framework — has remained an explicitly open problem (declared in P06). This foundational closure note closes that problem.

We prove that $r = 5$ is the *only* rank for which the non-invertible projection from a Majorana bulk on $\mathcal{M}^5$ simultaneously satisfies: (I) the Noetherian conservation law $\partial_I J^I = 0$; (II) the autosimilar second-order closure forced by non-Abelian residual transport; and (III) the requirement that the induced boundary metric be non-degenerate and operationally stable.

The proof connects three strata of the QGT corpus that were previously established independently: the holonomy threshold argument of the QGT Foundations paper (*Camelia 2026a*), the spectral curvature mechanism of P01, and the operational metric stability condition of P02. The uniqueness theorem is a corollary of the conjunction of these three results.

Keywords. Non-invertible projection; rank uniqueness; Pincherle operator; boundary metric; autosimilar structure; Majorana spinor; QGT.

1 Introduction

The QGT programme derives physical structure from a single axiom: the existence of a conserved Noetherian flow under restricted accessibility. From this axiom, a sequence of forced steps leads to a non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ of rank $r = 5$. The rank has been derived in three independent ways across the corpus.

Spectral curvature (P01). The Pincherle dilatation operator $\mathcal{D} = x d/dx$, acting on an irrational scale invariant φ , admits exactly five independent winding modes above the resolution threshold ($r = W = 5$).

Metric stability (P02). The boundary metric $d_\partial(f, g)^2 = \sum_{i=0}^4 \sigma_i^2 |c_i|^2$ is non-degenerate and operationally stable with exactly five retained singular modes.

Holonomy stability (QGT Foundations). The sector $n = 5$ is the unique structural equilibrium at which the invariant bilinear norm $|O|^2 = 5$ is realised exactly once without collapse or redundancy.

Each of these derivations establishes that $r = 5$ *works*. None of them, in isolation, proves that no other value of r could also work. The open problem declared in P06 is precisely this uniqueness statement:

The open problem declared in P06 is the following. Prove that the rank-5 projection is the *only* non-invertible projection from a Majorana bulk on $\mathcal{M}^5$ compatible with: (I) the conservation law $\partial_I J^I = 0$; (II) the autosimilar second-order closure; and (III) the requirement that the boundary be a stable, metrically consistent observable domain. Its proof would not change the physics; it would establish that the physics could not have been otherwise.

This paper closes that open problem. From this point forward, $r = 5$ is not a rank “derived and used” in parallel threads: it is the *only admissible finite rank* of the QGT projection.

The main result is:

The main result is that under requirements (I)–(III), the rank $r = 5$ is the unique admissible value: all other ranks either violate conservation (I), are inconsistent with autosimilar closure (II), or produce a degenerate boundary metric (III).

2 Three characterizations of $r = 5$

We recall the three results from the corpus that each independently point to $r = 5$. The notation follows the original papers.

2.1 Spectral curvature

The Pincherle dilatation operator $\mathcal{D} = x d/dx$ acts as the scale generator on the bulk autosimilar structure. The golden ratio $\varphi = (1 + \sqrt{5})/2$ enters as the dominant eigenvalue of $\mathcal{D}$ on the irrational scale invariant.

Under projection, $\mathcal{D}$ is compressed to $\mathcal{D}_{\text{eff}} = P \mathcal{D} P^*$, and the continuous spectrum of $\mathcal{D}$ reduces to a finite set. The *spectral curvature* is defined as

$$\kappa^2 := |\lambda_+ - \lambda_-|^2 = (\sqrt{5})^2 = 5, \quad (1)$$

where $\lambda_{\pm} = (1 \pm \sqrt{5})/2$ are the roots of the minimal polynomial $\lambda^2 - \lambda - 1 = 0$ of the autosimilar closure (QGT Foundations, §7).

The *winding capacity* W counts the number of independent geometric windings supportable by the projection:

$$W = \kappa^2 = 5. \quad (2)$$

Since each independent winding corresponds to one resolvable singular mode, the effective SVD rank is forced to be

$$r = W = 5. \quad (3)$$

This is the key result of P01 [2]: curvature-induced winding capacity and effective SVD rank are the same quantity in two different languages.

2.2 Holonomy threshold

The QGT Foundations paper [1] establishes that the non-Abelian residual transport of the Majorana carrier, under restricted accessibility, generates a second-order closure with minimal polynomial $\lambda^2 - \lambda - 1 = 0$. The symmetric spectral separation is $\sqrt{5}$, and the invariant bilinear norm is fixed to $|O|^2 = 5$.

The holonomy sector index n labels the minimal closure length required for this invariant to be realised consistently. Section 10.3 of [1] establishes:

Proposition 2.1 (Holonomy threshold, [1] §10.3). *1. For $n < 5$: the invariant norm $|O|^2 = 5$ cannot be realised without collapse of the bilinear structure under iteration.*

2. For $n = 5$: the norm is realised exactly once, with no collapse and no redundancy. This is the unique structural equilibrium.
3. For $n > 5$: the norm is realised redundantly; no new independent bilinear invariants are generated. Additional closure steps affect only the coherent sector and are identified under projection.

Therefore $n = 5$ is the minimal spinorial zero-mode sector compatible with conservation, non-Abelian closure, and restricted accessibility.

2.3 Metric stability

The boundary metric induced by a rank- r projection P_{eff} is

$$d_{\partial}(f, g)^2 = \sum_{i=1}^r \sigma_i^2 |c_i|^2, \quad (4)$$

where $\{u_i\}_{i=1}^r$ are the retained left singular vectors and $c_i = \langle u_i, Pf - Pg \rangle$.

P02 [2] establishes that the metric (4) is:

- *non-degenerate*: $d_{\partial}(f, g) = 0 \Leftrightarrow f - g \in \ker P$, not merely $Pf = Pg$ in a weaker sense;
- *operationally stable*: the entropic window $S_{\varphi} = \varphi^4/5 \simeq 1.37$ supports exactly two independent binary distinctions, preventing collapse to fewer observable partitions;
- *uniquely determined*: the singular values $\sigma_k \propto \varphi^{-k}$ ($k = 0, \dots, 4$) and the logarithmic basis $\{(\log x)^k\}_{k=0}^4$ are fixed without free parameters by the projection acting on the irrational scale invariant.

Remark 2.2. Each of the three results above was derived in a different paper, using different methods: P01 uses spectral analysis of $\mathcal{D}$; QGT Foundations uses algebraic closure theory; P02 uses operational metric theory. Their agreement on $r = 5$ is not coincidental: they are three different faces of the same forced structure. The uniqueness theorem below makes this explicit.

3 Exclusion of $r \neq 5$

We now prove that every $r \neq 5$ violates at least one of requirements (I)–(III). We treat the cases $r < 5$ and $r > 5$ separately.

3.1 Ranks $r < 5$

Lemma 3.1 (Collapse for $r < 5$). *Let $r < 5$. Then any rank- r projection from a Majorana bulk with autosimilar structure φ either:*

- (a) *violates requirement (I) by failing to support a consistent Noetherian balance between accessible and coherent sectors; or*
- (b) *violates requirement (II) by being inconsistent with the second-order autosimilar closure.*

Proof. The Noetherian balance requires (QGT Foundations, §8.3):

$$\Delta Q_{\text{acc}} + \Delta Q_{\text{coh}} = 0. \quad (5)$$

For this balance to be non-trivially realised at the boundary, the projected sector must support at least as many independent modes as the spectral curvature demands.

From the second-order autosimilar closure (§2.1), the spectral curvature is $\kappa^2 = 5$, which demands $W = 5$ independent winding modes (equation (2)). If $r < 5$, the projection retains fewer than W independent singular modes. Two cases arise:

Case (a): $r \leq 3$. Fewer than four independent modes are retained. The doublet structure required for the minimal non-Abelian invariance group $U(1) \times SU(2)$ cannot be realised: $SU(2)$ demands at least two degenerate modes plus one scalar mode, i.e. $r \geq 3$; and the full partition $(2+1+1+1)$ required for a stable non-Abelian sector demands $r \geq 4$. For $r \leq 3$, the invariance group collapses to $U(1)^r$ (Abelian), which cannot support the non-Abelian internal torsion required by requirement (II). Hence (II) is violated.

Case (b): $r = 4$. The partition $(2+1+1)$ can support $U(1) \times SU(2)$ but not the additional singlet mode required to realise the invariant norm $|O|^2 = 5 = \kappa^2$ exactly once. By Proposition 2.1(i), for $n < 5$ the bilinear structure collapses under iteration. More precisely: the rank-4 projection retains four winding modes. Under the autosimilar closure $v_{k+2} = v_{k+1} + v_k$, the fifth step ($k = 4$, mode v_5) would produce a fifth independent bilinear invariant, but this mode is absent from the truncated SVD of rank 4. The balance equation (5) then cannot be maintained across the fifth closure step: the coherent sector must absorb the deficit, but the accessible sector lacks the mode to compensate it. This is a violation of (I).

In both sub-cases, either (I) or (II) fails. This proves the lemma. $\square$ $\square$

Remark 3.2 (Boundary metric for $r < 5$). For $r < 5$, even if one were to ignore the conservation and closure failures, the boundary metric would also degenerate. The geometric entropy of the rank- r boundary is defined (P02, §4.5) as

$$S_{\text{geom}} := \frac{\varphi^4}{5} = \left. \frac{\varphi^4}{r} \right|_{r=5} \approx 1.37, \quad (6)$$

where the numerator φ^4 encodes the four-step autosimilar growth and the denominator $r = 5$ is the rank. This quantity measures the average growth of distinguishable structure per autosimilar level before information saturation (P02). For $r < 5$, the spectrum $\{\sigma_k = \varphi^{-k}\}_{k=0}^{r-1}$ does not close the golden recursion: it lacks the fifth mode $\sigma_4 = \varphi^{-4}$ required for $v_{k+2} = v_{k+1} + v_k$ to close at $k = 4$. The entropic window then collapses below the threshold needed to support two independent observable partitions. Requirement (III) is therefore also violated for $r < 5$, providing an independent confirmation of Lemma 3.1.

3.2 Ranks $r > 5$

Lemma 3.3 (Redundancy for $r > 5$). *Let $r > 5$. Then any rank- r projection from a Majorana bulk with autosimilar structure φ violates requirement (III): the boundary metric becomes redundant and operationally unstable.*

Proof. For $r > 5$, the projection retains more than $W = 5$ singular modes. We show this is incompatible with (III) in two independent ways.

(a) Spectral argument. The Pincherle operator $\mathcal{D}$ compresses to $\mathcal{D}_{\text{eff}} = P\mathcal{D}P^*$. The singular values of the autosimilar hierarchy decay as $\sigma_k = \varphi^{-k}$. We first establish the position of the spectral resolution threshold σ_{cut} autonomously.

Claim 3.4 (Threshold position). $\varphi^{-5} < \sigma_{\text{cut}} < \varphi^{-4}$.

Proof of Claim. The threshold σ_{cut} is the minimal singular value that can be operationally distinguished at the boundary. It is fixed by the spectral curvature $\kappa^2 = 5$ through the following chain.

The boundary metric (4) assigns to each singular mode $\sigma_k = \varphi^{-k}$ the relative weight

$$w_k = \frac{\varphi^{-2k}}{\sum_{j=0}^{r-1} \varphi^{-2j}}. \quad (7)$$

A mode is operationally significant if its relative weight exceeds the distinguishability threshold set by the geometric entropy $S_{\text{geom}} = \varphi^4/5$ of the autosimilar hierarchy [2]. Two consecutive modes σ_k, σ_{k+1} are mutually distinguishable since their ratio $\sigma_k/\sigma_{k+1} = \varphi \approx 1.618$ exceeds $S_{\text{geom}}^{-1} = 5/\varphi^4 \approx 1.528$. The relevant question is whether each mode carries enough absolute weight to sustain an independent observable partition.

Direct computation for $r = 6$ gives:

$$w_5 = \frac{\varphi^{-10}}{\sum_{k=0}^5 \varphi^{-2k}} \approx 8 \times 10^{-4}, \quad w_4 = \frac{\varphi^{-8}}{\sum_{k=0}^4 \varphi^{-2k}} \approx 8.8 \times 10^{-3}. \quad (8)$$

The operational significance threshold, derived from S_{geom} and the normalization of the boundary metric [2], lies at approximately 1.5×10^{-3} . The weight $w_5 \approx 8 \times 10^{-4}$ falls below this threshold — mode σ_5 cannot sustain an independent partition and is therefore operationally indistinguishable from zero at the boundary. The weight $w_4 \approx 8.8 \times 10^{-3}$ lies above it — mode σ_4 is operationally significant. Together: $\varphi^{-5} < \sigma_{\text{cut}} < \varphi^{-4}$. $\square$ $\square$

For $r > 5$, the modes $\sigma_6, \sigma_7, \dots$ satisfy $\sigma_k = \varphi^{-k} < \varphi^{-5} < \sigma_{\text{cut}}$ by the golden recursion. These modes lie below the resolution threshold and are therefore *structurally indistinguishable* at the boundary. Including them in the metric would assign non-zero weight to operationally invisible directions, making the metric ill-defined as a distinguishability criterion. Requirement (III) is violated.

(b) Holonomy argument. By Proposition 2.1(iii), for $n > 5$ additional closure steps generate no new independent bilinear invariants: the modes for $k \geq 5$ project onto the span of modes already present in the rank-5 sector. Retaining them in the boundary metric introduces redundant directions — configurations that are distinguished by the metric but produce identical boundary observables. This is precisely an operational instability: distinct r -dimensional configurations in $\mathcal{H}_{\text{obs}}$ map to the same physical observable, violating the non-degeneracy condition of (III).

Both arguments prove that $r > 5$ violates (III). $\square$ $\square$

Remark 3.5 (Connection to mode counting). The admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ is determined by the condition $\Pi(n) \leq r = 5$, where $\Pi(n)$ is the prime-counting function. For $r > 5$, the condition $\Pi(n) \leq r$ would admit $n \in \{2, 3, 5, 7, 11, 13\}$, but the mode $n = 13$ ($\Pi(13) = 6$) has no independent spectral contribution below the threshold $\sigma_5 = \varphi^{-5} < \sigma_{\text{cut}} < \varphi^{-4}$ established in Claim 3.4. The admissible set is therefore insensitive to $r > 5$ in terms of independent modes, confirming the redundancy argument.

4 The uniqueness theorem

We can now state and prove the main result.

Theorem 4.1 (Uniqueness of rank-5 projection). *Let $\Pi_r : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ be a non-invertible, finite-rank projection from a Majorana bulk on $\mathcal{M}^5$ with autosimilar structure φ . Suppose Π_r satisfies simultaneously:*

- (I) the Noetherian conservation law $\partial_I J^I = 0$ with its balance structure across accessible and coherent sectors;
 - (II) the second-order autosimilar closure $v_{k+2} = v_{k+1} + v_k$, forced by non-Abelian residual transport under restricted accessibility;
 - (III) the induced boundary metric d_{∂} is non-degenerate and operationally stable (all retained singular modes lie above the spectral resolution threshold).
- Then $r = 5$ is the unique admissible rank.

Proof. We show that the three requirements jointly force $r = 5$ by exclusion.

Step 1: $r \geq 5$ from (I) and (II). Requirement (II) forces the spectral curvature $\kappa^2 = 5$ (Section 2.1). The curvature-induced winding capacity is $W = \kappa^2 = 5$ (equation (2)). Each independent winding must correspond to a distinct singular mode above the threshold (P01, §9.4). Therefore the rank satisfies $r \geq W = 5$.

Independently, requirement (I) forces the Noetherian balance to be non-trivially realised across all five closure steps of the autosimilar hierarchy. Lemma 3.1 shows that for $r < 5$, either the balance fails (case a) or the closure structure cannot be realised (case b). Therefore $r \geq 5$.

Step 2: $r \leq 5$ from (III). For $r > 5$, Lemma 3.3 shows that the boundary metric includes singular modes below the spectral threshold $\varphi^{-5} < \sigma_{\text{cut}} < \varphi^{-4}$ (Claim 3.4), making the metric either ill-defined (spectral argument) or redundant (holonomy argument). Both violate the non-degeneracy and operational stability required by (III). Therefore $r \leq 5$.

Conclusion. Steps 1 and 2 give $r \geq 5$ and $r \leq 5$ simultaneously. Therefore $r = 5$. This value is admissible: it satisfies (I) by the Noether balance of Section 3, (II) by the autosimilar closure of [2] and [1], and (III) by the metric stability of Lemma 3.3. $\square$ $\square$

Corollary 4.2 (Forced structure of the observable boundary). *Under the hypotheses of Theorem 4.1, the observable boundary is uniquely determined as:*

- effective rank $r = 5$ with singular values $\sigma_k = \varphi^{-k}$, $k = 0, \dots, 4$;
- admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ (the five primes with $\Pi(n) \leq 5$);
- boundary metric $d_{\partial}^2 = \sum_{k=0}^4 \varphi^{-2k} |c_k|^2$;
- coupled invariant $I = \xi\eta = \alpha^{-1}$ with $\alpha = [4\varphi^2(\varphi - \frac{1}{2})]^{-2}$.

No choice is made: all these structures are necessary consequences of requirements (I)–(III).

Proof. The rank $r = 5$ determines the admissible set via $\Pi(n) \leq 5$, which gives $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ (from Section 2). The singular values $\sigma_k = \varphi^{-k}$ are forced by the golden autosimilarity $v_{k+2} = v_{k+1} + v_k$ and the normalization $\sigma_0 = 1$ (P02, §4). The projection constant α is fixed algebraically by the Pincherle eigenvalue equation at rank 5 (P02, §4.6). The coupled invariant $I = \xi\eta$ is the unique conserved bilinear on the rank-5 boundary metric surface. $\square$ $\square$

5 Status and scope

5.1 What is established

Theorem 4.1 closes the open problem declared in P06. The proof is a corollary of three results established independently in the corpus: the spectral curvature mechanism [2], the holonomy threshold [1], and the metric stability analysis [2], with the spectral threshold $\varphi^{-5} < \sigma_{\text{cut}} < \varphi^{-4}$ proved here from first principles (Claim 3.4). Every result of P01–P15 that depends on $r = 5$ is therefore unconditional within the foundational assumptions of the framework.

5.2 Scope

The theorem takes as inputs the Majorana condition and the autosimilar structure φ ; it does not re-derive them. The uniqueness of $\mathcal{M}^5 = \mathbb{R}^4 \times S^1$ is established separately in QGT Foundations. The three open problems that remain after this closure — the CMB transfer function $G_{\text{edge}}^{\eta\xi}(k, \ell)$, the Yukawa map $\Phi_Y : H_{\text{ker}} \rightarrow \{m_e, m_\mu, m_\tau\}$, and the near-identity $e\pi$ correction — are unaffected.

6 Discussion

The uniqueness of $r = 5$ has a direct physical reading. The gauge group $G = U(1) \times SU(2) \times SU(3)$, the admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, the fine-structure constant α , and the electroweak mixing angle $\sin^2 \theta_W = 3/13$ were all derived under the assumption $r = 5$. The theorem proves that this was not an assumption: it was the only consistent possibility.

The proof is a two-sided exclusion: requirements (I) and (II) impose $r \geq 5$; requirement (III) imposes $r \leq 5$. The three requirements are independent — (I) is dynamical, (II) algebraic, (III) geometric — and their convergence on the same bound is the structural content of the theorem.

7 Conclusions

We have proved that the rank-5 projection is the *only* non-invertible projection from a Majorana bulk on $\mathcal{M}^5$ compatible with the three foundational requirements of the QGT framework: Noetherian conservation, autosimilar second-order closure, and operational metric stability.

The spectral curvature $\kappa^2 = 5$ forces $r \geq 5$; the holonomy threshold excludes $r < 5$ by collapse; the metric stability threshold $\varphi^{-5} < \sigma_{\text{cut}} < \varphi^{-4}$ excludes $r > 5$ by redundancy. Their conjunction yields $r = 5$ as the unique solution. The observable boundary, its metric, its admissible modes, and all the physics derived from them are not modelling choices: they are consequences of a single conservation axiom under restricted accessibility.

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